

Sangwoo Hong

☎ +82-10-4169-9846 | ✉ swhong06@konkuk.ac.kr | 🏠 sangwoohong.github.io | 🎓 Google Scholar

Education & Experience

Konkuk University

Assistant Professor, Department of Computer Science and Engineering

- Teaching artificial intelligence courses for undergraduate students

Seoul, Korea

Mar. 2025 - present

Seoul National University

Ph.D. in Electrical and Computer Engineering

- Cognitive Machine Learning Lab (Advisor: Prof. Jungwoo Lee)

Seoul, Korea

Mar. 2020 - Aug. 2024

Seoul National University

B.S. in Electrical and Computer Engineering

- Advisor: Prof. Sunghyun Choi

Seoul, Korea

Mar. 2014 - Feb. 2020

Research Interests

My research interests lie in **Trustworthy AI**, with a focus on:

Safety	Adversarial Robustness, Verifiable Computing
Privacy	Data Unlearning, Memorization Mitigation
Fairness	Bias Mitigation, Spurious Correlation Mitigation
Alignment	Reward Modeling, Process Supervision
Interpretability	Mechanistic Interpretability, VLA Explainability

Publications

PredLM: A Sparse Large Language Model Decoding Accelerator with Hardware-Efficient Zero-Activation Prediction

Seunghyun Moon, Seowon Ji, **Sangwoo Hong**, and Eunji Kwon

IEEE Transactions on Circuits and Systems II: Express Briefs, 2026.

TCAS-II '26

Operator-aware shadow importance sampling for accurate fidelity estimation

Hyunho Cha, **Sangwoo Hong**, and Jungwoo Lee

Physical Review A, 2026.

PRA '26

MVP-LAM: Learning Action-Centric Latent Action via Cross-Viewpoint Reconstruction

Jung Min Lee, Dohyeok Lee, Seokhun Ju, Taehyun Cho, Jinwoo Koo, Li Zhao, **Sangwoo Hong**, and

Jungwoo Lee

Forty-Third International Conference on Machine Learning, 2026.

ICML '26

Efficient Process Reward Modeling via Contrastive Mutual Information

Nakyung Lee, **Sangwoo Hong**, and Jungwoo Lee

The 64th Annual Meeting of the Association for Computational Linguistics, 2026

ACL Main '26

Bias Alleviation through Network Pruning for Sparse and Debiased Models

Sangwoo Hong, Sehwan Kim, Hyungjun Joo, Hyeonggeun Han, Jiyeon Shin, Yoav Wald, and Jungwoo

Lee

IEEE Transactions on Image Processing, 2026

TIP '26

An Adaptive Sampling Framework for Diffusion-based Dataset Distillation with High Fidelity and Diversity

Sunbeom Jeong, Sehwan Kim, Hyeonggeun Han, Hyungjoon Joo, **Sangwoo Hong**, and Jungwoo Lee

The 40th Annual AAAI Conference on Artificial Intelligence

AAAI '26 Oral

Adjusting Initial Noise to Mitigate Memorization in Text-to-Image Diffusion Models

Hyeonggeun Han, Sehwan Kim, Hyungjun Joo, **Sangwoo Hong**, and Jungwoo Lee

Conference on Neural Information Processing Systems

NeurIPS '25

Constructing Fair Latent Space for Intersection of Fairness and Explainability

Hyungjoon Joo, Hyeonggeun Han, Sehwan Kim, **Sangwoo Hong**, and Jungwoo Lee

The 39th Annual AAAI Conference on Artificial Intelligence

AAAI '25

Mitigating Spurious Correlations via Disagreement Probability

Hyeonggeun Han, Sehwan Kim, Hyungjun Joo, **Sangwoo Hong**, and Jungwoo Lee

Conference on Neural Information Processing Systems

NeurIPS '24

Group-wise Verifiable Coded Computing under Byzantine Attacks and Stragglers

Sangwoo Hong, Heecheol Yang, Youngseok Yoon, and Jungwoo Lee

IEEE Transactions on Information Forensics and Security (JCR: 6.39%)

TIFS '24

Straggler-Exploiting Fully Private Distributed Matrix Multiplication With Chebyshev Polynomials

Sangwoo Hong, Heecheol Yang, Youngseok Yoon, and Jungwoo Lee

IEEE Transactions on Communications (JCR: 7.53%)

TCOM '23

Hierarchical Group Testing for Byzantine Attack Identification in Distributed Matrix Multiplication

Sangwoo Hong, Heecheol Yang, and Jungwoo Lee

IEEE Journal on Selected Areas in Communications (JCR: 1.6%)

JSAC '22

Chebyshev Polynomial Codes: Task Entanglement-based Coding for Distributed Matrix Multiplication

Sangwoo Hong, Heecheol Yang, Youngseok Yoon, and Jungwoo Lee

Proceedings of the 38th International Conference on Machine Learning

ICML '21

Byzantine attack identification in distributed matrix multiplication via locally testable codes

Sangwoo Hong, Heecheol Yang, and Jungwoo Lee

IEEE International Symposium on Information Theory

ISIT '22

Private and secure coded computation in straggler-exploiting distributed matrix multiplication

Heecheol Yang, **Sangwoo Hong**, and Jungwoo Lee

IEEE International Symposium on Information Theory

ISIT '21

Manuscripts Under Review

Look Where You Drive: Action-Based Visual Explanation for Autonomous Driving VLAs

Dohyoung Kim, Donghyun Kim, Jaehyuk Yu, and **Sangwoo Hong**

Under Review

Mechanistic Circuit Identification for Controllable Data Generation

Nakyung Lee, **Sangwoo Hong**, and Jungwoo Lee

Under Review

Professional Services

Area Chair, Conference on Neural Information Processing Systems

NeurIPS 2026

Session Chair, Association for Computational Linguistics

ACL 2026

Session: Interpretability and Analysis of Models for NLP

Organizing Committee, International Symposium on Information Theory

ISIT 2028

Reviewer for Journals and Conferences

NeurIPS, ICML, AAAI, CVPR, ICCV, ECCV, TNNLS, TIFS, TCOM, ACL